BharatTrip Refund Escalations: Business Case & Solution Proposal

Executive Summary (TL;DR)
BharatTrip is experiencing a spike in customer/agent refund escalations despite stable refund volumes. Our analysis identified a severe operational disconnect between Support and Finance, leading to **100 dropped tickets** and **149 short payments**. By implementing a **Single Source of Truth (SSOT)** and an **AI-Driven Data Reconciliation Workflow**, we deploy a "Digital Workforce." This allows existing teams to eliminate manual data leakage and automate agent communications, driving a projected **85%+ drop in escalations** over the next quarter **without adding a single headcount**.

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1. Defining the Problem

Customer and agent escalations regarding refunds are rising. The core underlying problem is a **severe operational disconnect between the Support and Finance teams**, characterized by siloed tracking systems and inconsistent definitions of refund statuses.

Key Communication Failures:

1. **Data Leakage**: Informal requests via WhatsApp and Email bypass the tracker altogether, meaning they are never reviewed or processed.
2. **"Closed" Status Ambiguity**: The Support team marks tickets as "Closed" once handed off to Finance, misleading the agent into believing the refund has been disbursed.
3. **Hidden Policy Deductions**: Finance applies necessary deductions (e.g., cancellation fees) and processes payouts, but this information is never relayed back to the agent by Support, causing "short payment" escalations.

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2. Data Analysis & Insights

By cross-referencing the `Support_Tracker`, `Finance_Tracker`, and `Escalations` logs via a deep database join, we identified the specific patterns driving the **155 total escalations**.

The Discrepancy Breakdown

Escalation Driver	Ticket Volume	Description
Dropped Handoffs	**100**	Logged by Support, but entirely missing from Finance tracking.
Deduction Mismatches	**149**	Amount promised by Support differs from amount disbursed by Finance.
Off-Tracker Leakage	**24**	Escalations with "no ref" originating from unlogged WhatsApp/Email.
Asynchronous Closure	**47**	Processed by Finance but missing from Support.

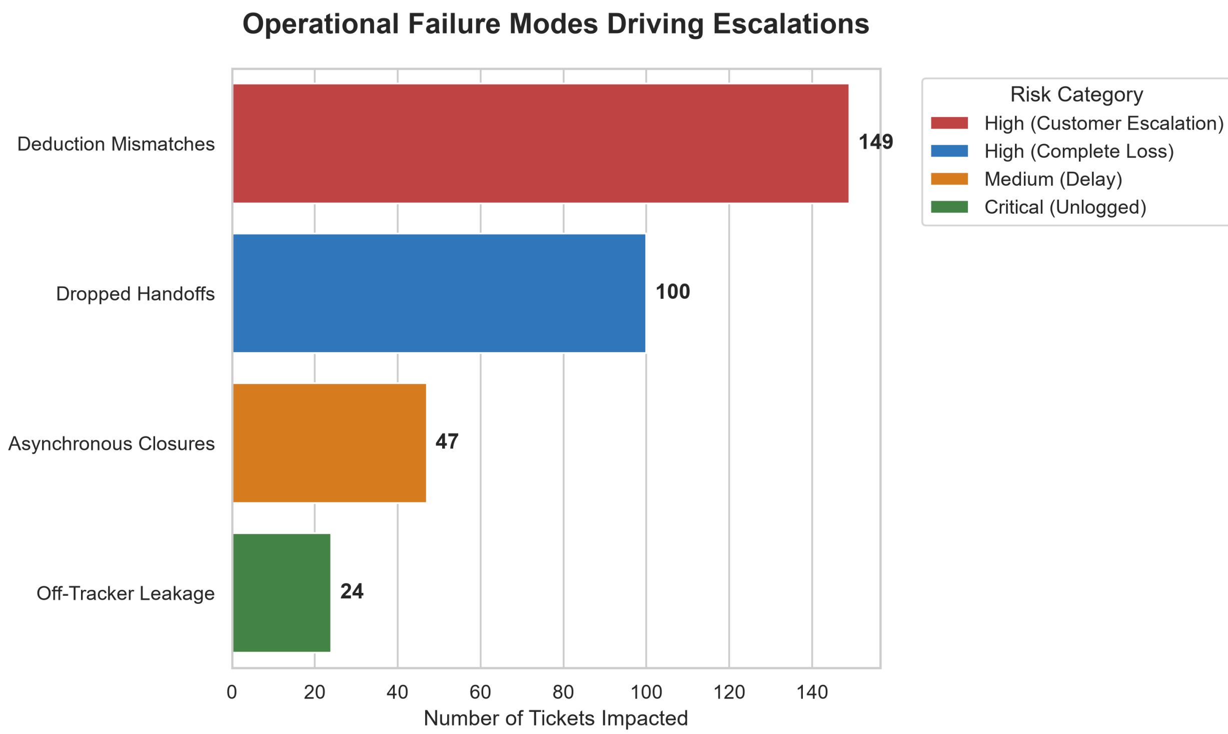

Figure 1: Visual breakdown of current support processing inefficiencies highlighting the 100 dropped handoffs and 149 mismatches.

> **Insight:** While a naive net volume calculation (755 Support - 689 Finance) suggests only 66 missing tickets, our deep ID-to-ID database reconciliation reveals the true scale of the failure is **100 dropped tickets**, masked by duplicate records and asynchronous closures.

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3. Designing a Zero-Headcount Solution

To permanently bring escalations down without adding headcount, BharatTrip must establish a unified **Single Source of Truth (SSOT)** and automate the communication loop.

The Proposed Workflow (Digital Workforce):

1. **Unified SSOT Tracking**: Migrate both teams onto a single shared database.
2. **AI-Powered Ingestion**: Deploy an AI agent to monitor Support Email and WhatsApp. The AI acts as a digital clerk, parsing unstructured messages and logging them as structured tickets, eliminating "no ref" leakage entirely.
3. **Automated Reconciliation**: A system trigger automatically cross-references Finance payouts with Support quotes. If a deduction occurs, the AI drafts an explanatory email for the agent, preventing short-payment shock.

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4. The AI Prototype: Enterprise-Ready Architecture

To deliver a production-ready solution, I developed a **Streamlit Web Application** showcasing an advanced Multi-Agent architecture powered by `gemini-3.5-flash`. Rather than building a simple LLM script, I engineered a secure, deployable enterprise tool.

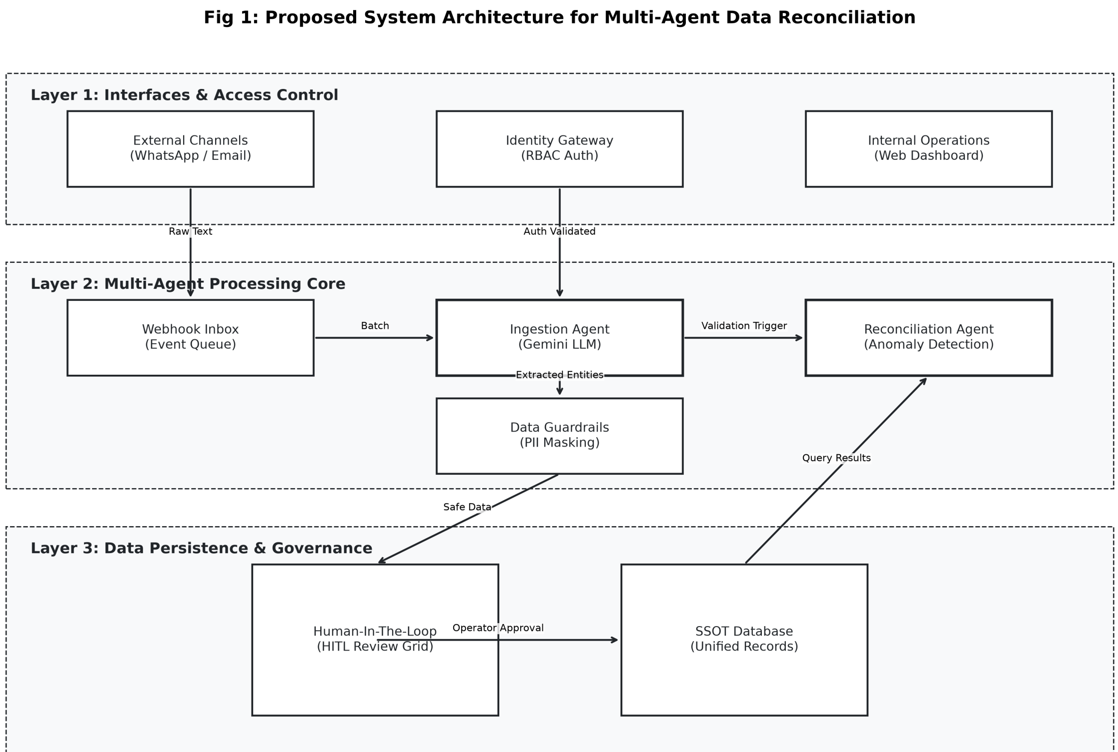

Figure 2: Proposed Multi-Agent System Architecture Flowchart

Key Enterprise Features & Security Mitigations

Deploying AI in financial workflows carries inherent risks (e.g., hallucinated refund amounts, unauthorized access, or PII leaks). Our architecture mitigates this using a **Defense-in-Depth** approach:

- **Identity Gateway & RBAC**: The application enforces dual-role authorization. Managers have full access, while Operators are restricted.
- **Data Privacy (PII Redaction)**: Sensitive identifiers in unstructured messages are automatically redacted via RegEx prior to ingestion. Furthermore, Operator roles see dynamically masked UI views.
- **Data Loss Prevention (DLP)**: UI-level CSS injection prevents text selection and copying of sensitive fields.
- **AI Confidence Guardrails**: The ingestion agent scores its own extractions and flags low-confidence data for mandatory human review.
- **Human-In-The-Loop (HITL) & Auditability**: The AI is strictly limited to *drafting* emails and structuring data. A human must click "Approve & Send", and every action is logged to a secure CSV for financial compliance.
- **Unified Database Explorer**: A global search interface completely breaks down the silos between Support and Finance, serving as the ultimate Single Source of Truth.

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5. Measuring Success

To ensure the solution is working, we will monitor two primary metrics over the next month:

1. **Escalation Rate**: The percentage of refund requests that result in a formal escalation. **Target: 85%+ drop within 30 days.** Because 149 mismatches + 100 drops + 24 WhatsApp leaks account for virtually all operational friction, successfully deploying this SSOT proactively eliminates the root causes of the vast majority of escalations.
2. **Tracker Discrepancy Count**: The number of tickets marked "Closed" by Support without a matching payout record in Finance. **Target: Zero.**

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6. Assumptions Made

As the brief was deliberately incomplete, I proceeded with the following assumptions to build the prototype:

- **Finance is the ultimate authority on payout amounts**: If Support quotes 5000 INR but Finance pays 4500 INR, I assumed the 500 INR deduction is legitimate (cancellation fee) rather than a Finance error, meaning the fix requires better communication to the agent rather than a refund correction.
- **WhatsApp/Email formats are unpredictable**: I assumed agents will not follow a strict template, necessitating an LLM to dynamically extract the `agent_name`, `route`, and intent from informal natural language.